Annex 10

Regulatory considerations for the life cycle of medicinal oxygen

For any further information or request, please send an email to the Norms and Standards for Pharmaceuticals (NSP) Team at WHO, at email nsp@who.int.

Contents

Abbreviations

GMP	good manufacturing practices
GSDP	good storage and distribution practices
NRA	national regulatory authority
PSA	pressure swing adsorption
QMS	quality management system
VPSA	vacuum pressure swing adsorption
VSA	vacuum swing adsorption
WHO	World Health Organization

1. Introduction

Medicinal oxygen is a lifesaving and critical therapeutic gas, with no substitutions in the treatment and prevention of hypoxaemia and hypoxia. It plays a vital role in a wide range of clinical settings, from intensive care and emergency medicine to home-based therapy. Its use is especially important in the care of newborns, children with pneumonia, surgical and trauma patients, and vulnerable populations, such as elderly people and pregnant women. Recognizing its essential role, the World Health Organization (WHO) includes medicinal oxygen in its Model List of Essential Medicines (*1*), emphasizing that it has no effective substitute in the treatment of respiratory failure (*2*).

Oxygen for use in health care facilities[40] can be obtained in several forms: medical liquid oxygen, produced through cryogenic distillation and stored in cryogenic tanks before being vaporized for use; compressed medicinal oxygen, filled into high-pressure cylinders and connected to manifold systems; and on-site oxygen production using generator plants, such as pressure swing adsorption (PSA), vacuum swing adsorption (VSA), or vacuum pressure swing adsorption (VPSA), which extract oxygen from ambient air (*3*). Each source requires suitable infrastructure, reliable power and trained personnel to ensure safe and continuous delivery of oxygen to patients.

When intended for therapeutic, prophylactic (prevention of hypoxia during and after surgery by maintaining oxygen saturation and prevention of altitude sickness above 3000 meters), diagnostic, or other medical purposes, medicinal oxygen is regulated as a medicinal product. As such, it must meet the

[40] Includes hospitals, clinics or similar facilities that provide patients with their health care needs.

same minimum standards for quality, safety and efficacy as other pharmaceutical products. Medicinal oxygen achieves its therapeutic effect primarily through a pharmacological mechanism – by increasing the partial pressure of oxygen in blood and tissues. To ensure patient safety, its production, handling and distribution must comply with good manufacturing practices (GMP) and be free from contaminants such as water, carbon monoxide, sulfur dioxide, hydrocarbons and particulates.

Oxygen is available in the market for industrial and medical uses, but these are not interchangeable. Medicinal oxygen is different from industrial oxygen in purity and quality, which is influenced by a robust manufacturing process and testing controls. Medicinal oxygen must be manufactured under GMP conditions and tested for compliance with pharmacopoeial specifications, including identity, purity and assay. In contrast, Industrial oxygen is not subject to the same purity, testing or contamination controls, and may contain harmful substances such as hydrocarbons, particulates or other industrial residues, making it unsafe for medical use. Regulatory systems must enforce strict segregation in production, sourcing, labelling and distribution to prevent the inadvertent or inappropriate use of industrial-grade oxygen for medical purposes.

Medicinal oxygen can be produced through two main methods: large-scale industrial manufacturing and on-site generation. Large-scale production often employs cryogenic distillation of ambient air to achieve high-purity liquid oxygen, which is then stored and distributed via cylinders or cryogenic tanks, and compressed medicinal oxygen, produced by compressing liquid oxygen and vaporizing it to produce a high-pressure gas, which is then filled into high-pressure gas cylinders. On-site generation, typically via oxygen generator plants such as PSA technology, produces the gas by removing the nitrogen from ambient air using molecular sieves. It provides a reliable source of oxygen directly at health care facilities. While PSA systems are predominantly used for immediate patient care, they can also be employed for cylinder filling, provided that appropriate quality and safety standards are met. National regulatory frameworks should not unduly restrict such practices when adequate oversight is in place.

Despite its classification as an essential medicine, regulatory oversight of medicinal oxygen remains inconsistent globally. A WHO desk review revealed that only 80 out of 194 Member States currently regulate medicinal oxygen as a medicinal product. In many low- and middle-income countries, national regulatory authorities (NRAs) either lack legal frameworks for oxygen or do not actively enforce them. This regulatory gap poses serious risks to patient safety and contributes to unreliable oxygen access in critical settings.

To address these gaps, WHO has developed a regulatory consideration document covering the production, distribution, storage and delivery of medicinal oxygen. This document complements existing guidance, including WHO publications, *The International Pharmacopoeia* (4), and other relevant resources, to support the development of robust regulatory systems for medicinal oxygen.

A structured approach is essential for effective regulation. NRAs are encouraged to carry out gap analyses in collaboration with relevant stakeholders, integrating best practices from established regulatory systems and adapting them to their local context. This will help identify key priorities and requirements. Developing detailed implementation plans, which include clear transition periods, action roadmaps, timelines and deliverables, will be critical for ensuring regulatory compliance and securing a consistent and reliable supply of medicinal oxygen. Whereas medicinal oxygen intended for administration to a patient is regulated as a medicinal product, the associated administration equipment may be regulated as a medical device, depending on the equipment.

2. Purposes

This regulatory consideration document provides recommendations for NRAs on the oversight of medicinal oxygen throughout its life cycle, covering production, quality control, storage, distribution and post-market activities. It also aims to support manufacturers, importers and distributors in understanding the regulatory requirements needed to ensure that medicinal oxygen consistently meets established quality and safety standards and is effectively delivered to patients.

3. Scope

The document outlines a regulatory framework for the entire life cycle of medicinal oxygen, including pre-market activities such as production and quality control, as well as post-market activities such as market surveillance and vigilance. It focuses on medicinal oxygen and its primary packaging, including container closures and valves. Permanently affixed valves or integrated pressure regulator assemblies that are not intended for removal are considered part of the authorized medicinal product and fall within the scope of this guidance.

Also, the document outlines recommendations for medicinal oxygen produced on a large scale for commercial distribution and for oxygen manufactured or handled at health care facilities for use with their patients.

It does not cover regulatory requirements for medicinal oxygen produced at home for personal use. Nonetheless, the principles outlined here may be applied in such contexts to ensure that home-generated oxygen is suitable for its intended use and meets appropriate quality standards. In addition, the recommendations in this document exclude oxygen manufactured for industrial purposes.

4. Glossary

active substance gas. Any gas intended to be an active substance for a medical product or medicinal gas (*5*).

authorized person. The person recognized by the national regulatory authority as having the responsibility for ensuring that medicinal oxygen made available to the end user has been manufactured, tested and approved for release in compliance with the laws and regulations in force in that country (*6*).

distribution. Procuring, purchasing, holding, storing, selling, supplying, importing, exporting or movement of pharmaceutical products, with the exception of dispensing or providing pharmaceutical products directly to a patient or their agent (*7*).

distributor. Any natural or legal person in the supply chain who, on their own behalf, furthers the availability of medicinal oxygen to the end user (*8*).

importation. The act of bringing or causing to be brought any goods into a customs territory (national territory, excluding any free zone) (*7*).

importer. An individual or company or similar legal entity importing or seeking to import a medical product. A "licensed" or "registered" importer is one who has been granted a licence for the purpose (*9*).

manifold. Equipment or apparatus designed to enable one or more gas containers to be emptied or filled at the same time (*10*).

manufacturer. A company that carries out operations such as production, packaging, repackaging, labelling and relabelling of pharmaceuticals (*5*).

manufacturing. Manufacture consists of the operations of division or distribution and filling into dedicated pharmaceutical and medicinal packaging. Packaging is often automated and may include prior modification of the physical state of the gas (vaporization by heating of liquefied gas and compression) (*7*).

market surveillance. The activity carried out by authorities to ensure that products on the market conform to the applicable laws and regulations and comply with existing requirements. Market surveillance covers a full range of actions, including the monitoring and control of the market and, where necessary, the imposition of corrective measures and penalties. It involves close contact of authorities with stakeholders, consumers and consumer organizations (*11*).

medical device. Any instrument, apparatus, implement, machine, appliance, implant, reagent for in vitro use, software, material or other similar or related article intended by the manufacturer to be used, alone or in combination, for human beings for one or more of the specific medical purposes of:

- diagnosis, prevention, monitoring, treatment, or alleviation of disease;
- diagnosis, monitoring, treatment, alleviation of, or compensation for an injury;
- investigation, replacement, modification or support of the anatomy or physiological process;
- supporting or sustaining life;
- control of conception;
- cleaning, disinfection, or sterilization of medical devices;
- providing information by means of in vitro examination of specimens derived from the human body;
- that which does not achieve its primary intended action by pharmacological, immunological or metabolic means, in or on the human body, but which may be assisted in its intended function by such means (*8*).

medicinal oxygen.[41] Medicinal oxygen is an essential medicine used for therapeutic purposes. It is defined as oxygen that meets the purity standards set forth by a pharmacopoeia (for example, *The International Pharmacopoeia*, *United States Pharmacopeia*, or *European Pharmacopoeia*) and is intended for human use in medical applications. Medicinal oxygen gas is a finished product that has undergone all stages of production and quality control, including packaging in its final container and labelling. Medicinal oxygen gas product is released to the market by an authorized person and can be used directly by health care professionals or by the patients themselves. Medicinal oxygen can be presented as a medicine, but also as an active substance and as a service product from an oxygen concentrator.

national regulatory authority. A government body or other entity that exercises a legal right to control the use or sale of medical products within its jurisdiction, and that may take enforcement action to ensure that medical products marketed within its jurisdiction comply with legal requirements (*8*).

post-market surveillance for medicines. The science and activities related to the detection, assessment, understanding, and prevention of adverse effects

[41] In some jurisdictions, medicinal oxygen is used interchangeably with medical oxygen. For the purpose of this document, both terms refer to the same product.

or any other medicine- or vaccine-related problem. Essentially, post-market surveillance ensures the safe use of medicines by monitoring their effects throughout their life cycle (*12*).

production. All operations involved in the preparation of a pharmaceutical product, from receipt of materials, through processing, testing, packaging, and repackaging, labelling, and relabelling, to completion of the finished product (*13*).

purity standard. Medicinal oxygen must meet the purity requirements specified by the relevant pharmacopoeia. For example, at least 99.5% by cryogenic distillation or oxygen 93% by oxygen generator plants (which contains not less than 90% and not more than 96% volume in volume (v/v) of oxygen, the remainder consisting mainly of argon and nitrogen) (*4*).

5. Life cycle of medicinal oxygen from production to patient

Regulatory considerations include understanding the linked processes and responsibilities that underpin the effective management of medicinal oxygen throughout its life cycle.

Table A10.1 provides an overview of the key stages in the life cycle of medicinal oxygen, from production to patient administration. It highlights the critical activities, stakeholders involved, and relevant considerations at each stage to ensure the quality and purity of medicinal oxygen.

Table A10.1

Stages in the medicinal oxygen life cycle: from production to patient administration

Life cycle	Who?	Legislation	Responsibility	How	Examples of standards and norms	Role of the regulator
Manufacturing						
Active substance gas supplier	Manufacturer	Medicine law	Manufacturer	Cryogenic distillation	GMP for medicinal gases (5, 6)	✓ Inspection
Manufacturing site	Manufacturer	Medicine law	Manufacturer	QMS, GMP	✓ Manufacture authorization ✓ Licensing ✓ Inspection (1, 2, 10)	
Health care facility	Designated personnel	Health policy Medicine law	Pharmacist	Production via oxygen generator plants: PSA, VSA, VPSA	GMP (1, 2) GSDP (13)	✓ Licensing ✓ Inspection
Oxygen distribution						
Distribution, import	Manufacturer, distributor	Medicine law	Manufacturer, importer or distributor	Tanks and tankers, cylinders	GSDP (13) ISO	✓ Market authorization ✓ Inspection ✓ Licensing

Table A10.1 *continued*

Life cycle	Who?	Legislation	Responsibility	How	Examples of standards and norms	Role of the regulator
Health care facility						
Oxygen delivery	Pharmacist	Medicine law Medical devices regulation	Pharmacist, engineer, nurses	Cylinders, medical gas pipeline system, pressure valves, nozzles	WHO guidance (14, 15) ISO	✓ Inspection ✓ Surveillance
Receiving and storage	Pharmacist	Medicines law	Pharmacist	Tanks, cylinder	WHO guidance (15) ISO	✓ Inspection
Oxygen administration to patient	Clinicians, nurses, anaesthetists, respiratory therapists	Medicine law Health policy Clinical practice standards	Prescriber and health care provider	Administered via nasal cannula, face mask, non-rebreather mask, or ventilator	WHO guidance (14, 15) ISO	✓ Inspection of clinical practices ✓ Pharmacovigilance ✓ Clinical surveillance

GMP = good manufacturing practices; GSDP = good storage and distribution practices; ISO = International Organization for Standardization; QMS = quality management system.

Note: Examples of applicable ISO standards for medicinal oxygen regulation and use include the following.

- ISO 15001:2010. Anaesthetic and respiratory equipment – Compatibility with oxygen.
- ISO 8573-1:2010. Compressed air – Part 1: Contaminants and purity classes.
- ISO 7396-1:2016. Medical gas pipeline systems – Part 1: Pipeline systems for compressed medical gases and vacuum.
- ISO 10156:2017. Gas mixtures – Determination of fire potential and oxidizing ability for the selection of cylinder valve outlets.
- ISO 20421-1:2018. Cryogenic vessels – Large transportable vacuum-insulated vessels – Part 1: Design, fabrication, inspection and testing.
- ISO 10524-1:2018. Pressure regulators for use with medical gases – Part 1: Pressure regulators and pressure regulators integrated with flow-metering devices.
- ISO/TR 24971:2020. Medical devices – Guidance on the application of ISO 14971.

6. Legal and regulatory considerations for medicinal oxygen

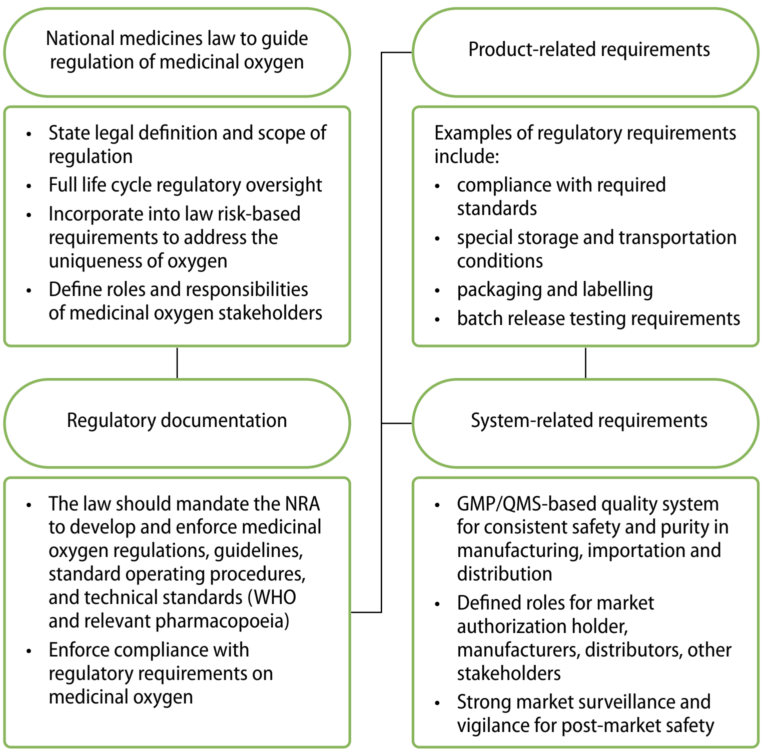
The regulatory framework for medicinal oxygen should follow a similar approach to that of medicines, ensuring comprehensive regulation of the entire life cycle of the product from development and approval to market entry and patient availability. Although medicinal oxygen shares many characteristics with medicines, it also presents unique considerations (such as its physical form, production process and delivery systems) that may require adaptations within regulatory frameworks. National law should guide regulation of medicinal oxygen and, in this context, national medicines legislation should form the basis for regulation. Where not already established, national frameworks may consider explicitly classifying medicinal oxygen as a medicinal product under pharmaceutical legislation to ensure appropriate regulatory oversight. The law should empower the NRA to license establishments and oversee all aspects of medicinal oxygen, including initial authorization, renewal, and ongoing compliance monitoring through inspections and reporting mechanisms. The NRA should have the legal power and resources to ensure that all medicinal oxygen produced meets minimum quality assurance standards and complies with pharmacopoeial requirements (for example, *The International Pharmacopoeia*, *United States Pharmacopeia*, or *European Pharmacopoeia*) for quality and purity. The ministry of health must prioritize investing in and strengthening NRA capacity, rather than substituting its role in regulating medicinal oxygen. It is important that medicinal oxygen be clearly defined within law or regulations to establish its regulatory status and ensure appropriate application of pharmaceutical oversight mechanisms. The definition should differentiate between industrial or non-medicinal oxygen and the oxygen intended for medical purposes, emphasizing the intended use of the latter. It should also encompass aspects such as purity, quality, delivery systems to health care facilities, and proper storage conditions. The legal definition may also refer to specifications outlined in relevant pharmacopoeias and WHO guidance to ensure alignment with internationally recognized standards.

High-level requirements should be included within the laws and regulations, ensuring they align with the guiding principles of applicable legislation. The law should prioritize safety, quality, purity and oversight, treating medicinal oxygen with the same level of care and compliance with specifications as medicines. Given its critical role in patient care, regulation of medicinal oxygen should address its unique characteristics, including its associated risk of fire, specific handling requirements, and the risks posed by impurities, through a structured risk-based regulatory approach that informs oversight, inspection and mitigation measures. Additionally, because medicinal oxygen is often delivered through specialized equipment such as oxygen concentrators and delivery systems, which are regulated as medical devices, and through pressure equipment such as cylinders, the regulatory framework should include requirements to ensure these components meet applicable safety standards, undergo regular inspections, and function properly to prevent failures and adverse events (Fig. A10.1).

Medicinal oxygen regulations should address, at a minimum, the following:

- regulatory requirements for production facilities and equipment;
- quality control processes, including testing for purity and contamination;
- proper labelling and packaging to guarantee safe handling and use;
- regulatory oversight procedures, including inspection and approval mechanisms;
- post-market surveillance.

Fig. A10.1
Legal and regulatory framework for medicinal oxygen



The law should require the licensing of manufacturers, importers and distributors interested in engaging in production, importation, exportation or distribution of medicinal oxygen. The legal framework should also define the roles and responsibilities of all stakeholders, including NRAs, manufacturers, importers, distributors and health care facilities when medicinal oxygen is manufactured on site. It should establish a structure that facilitates collaboration across multiple agencies, ensuring that various aspects of medicinal oxygen regulation, such as purity, safety and quality, are effectively managed and monitored by relevant authorities. These guidelines should be regularly reviewed and updated to align with WHO and other international standards, as applicable. In addition, detailed guidelines should be provided to stakeholders to clarify aspects of medicinal oxygen, such as production, storage, handling and monitoring. The guidelines should address the regulatory components of both pre-market and post-market authorization and logistical aspects. They should be reviewed regularly and aligned with international best practices, as applicable.

It is recommended that the NRA oversee and conduct post-market surveillance activities, thus ensuring that manufacturers adhere to the necessary regulations and that proper mechanisms for monitoring, sampling and testing, reporting, and addressing adverse events are in place. Additionally, the legal framework should include provisions to authorize the NRA to adopt the principles of reliance and recognition pathways and leverage decisions, including regulatory decisions on medicinal oxygen made by competent authorities or institutions in other jurisdictions, where appropriate. While reliance can facilitate faster access to medicinal oxygen and support efficient use of resources, the NRA should nonetheless retain full accountability for all final regulatory decisions.

Note: The regulatory authority is expected to carry out all core functions to ensure the effective regulation of medicinal oxygen, including issuance of licences; issuance of marketing authorization; regulatory inspections defining the roles and responsibilities of relevant stakeholders; strengthening interagency collaboration; providing clear regulatory guidelines to guide regulation of the entire life cycle of medicinal oxygen; implementing post-market surveillance activities; and adopting principles of reliance and recognition to streamline regulatory processes.

6.1 Role and responsibilities of the national regulatory authority

The NRA should have functions and powers to implement, enforce and regulate all matters relating to medicinal oxygen in compliance with good regulatory practices (*16*). The NRA shall ensure the safety, quality and efficacy of the medicinal oxygen through authorization and inspection activities. This should be achieved through evaluation of scientific data, on-site inspections, and laboratory testing on medicinal oxygen prior to market authorization. Table A10.2 outlines the key roles and responsibilities of NRAs in regulating medicinal oxygen.

Table A10.2
Roles and responsibilities of the NRA in medicinal oxygen regulation

Regulatory function	Key responsibilities and activities
Market authorization	• Define medicinal oxygen as a medicinal product, detailing its specific characteristics, such as form, purity, production process, distribution pathways and appropriate storage. • Outline the regulatory principles applicable to medicinal oxygen within the broader context of medicine regulation. • Align national regulatory requirements, such as those recommended by WHO, to ensure consistency, facilitate global market access, and promote reliance and recognition of regulatory decisions across countries and regions. • Publish the decision on medicinal oxygen as a medicinal product. • Publish technical dossier requirements for medicinal oxygen. • Establish or adopt internationally harmonized specifications for medicinal oxygen. • Establish requirements for market authorization of medicinal oxygen applications and licensing of establishments. • Establish requirements for medicinal oxygen labels, labelling and packaging. • Develop guidance and guidelines to assist the industry to comply with the regulatory requirements.
Inspection and enforcement	• Describe the key steps involved in the production, quality control and distribution of medicinal oxygen, highlighting the associated regulatory requirements at each stage. • Establish or adapt internationally harmonized specifications for GMP and good storage and distribution practices (GSDP) for medicinal oxygen. • Establish requirements for storage and transportation and ensure cylinders and piping in storage and distribution of medicinal oxygen meet the applicable standards. • Conduct regular inspections to ensure ongoing regulatory compliance. • Conduct on-site inspection of manufacturing sites and of importers and distributors.

Table A10.2 *continued*

Regulatory function	Key responsibilities and activities
Post-market surveillance	• Ensure robust mechanisms for post-market surveillance to monitor the product's safety, quality and efficacy. • Designate testing laboratories and set regulatory requirements to ensure their effective functioning. • Establish systems for tracking and reporting post-market surveillance and vigilance. • Establish requirements for reporting and documentation of adverse events. • Establish requirements for handling complaints, including reporting compliance status and any deviations to NRAs, and recalls.

6.1.1 Requirements for market authorization

Due to its peculiarity, medicinal oxygen may be presented either as a finished medicinal product or as an active substance. It is recommended that the NRA grant marketing authorization to manufacturers of medicinal oxygen to ensure their legal accountability for the product's quality, safety and efficacy throughout its entire life cycle. The regulatory framework should set out clear requirements that manufacturers must comply with before bringing medicinal oxygen to market. The market authorization process should include requirements related to registration of medicinal oxygen.

Applications submitted for market authorization of medicinal oxygen should contain, at a minimum, information describing the manufacturing process, its control, packaging, the specification of medicinal oxygen to be authorized, the name and address of the applicant, the name and address of the facility or facilities where the medicinal oxygen is or will be manufactured, and any other information as prescribed by the NRA. It is the responsibility of applicants to provide necessary information in accordance with the regulatory requirements to allow the NRA to make regulatory decisions concerning market authorization. The NRA should establish clear guidelines for the types of authorizations required for medicinal oxygen, ensuring the quality, safety and efficacy of the product.

Documents required for market authorization

Documentation requirements for market authorization should contain the following (*10*).

- Information and description of physical chemical characteristics of the active substance gas.

- Description of the product, including the name of the gas, physical form, name and address of the manufacturer, pressure and concentration, packaging (for example, cylinder, mobile cryogenic container or fixed cryogenic container).
- Data demonstrating the stability of medicinal oxygen during the various stages of the product life cycle as a function of its conditions of storage, transport, and handling (may be bibliographic data), or justification for not performing new studies, supported by literature references or pharmacopoeial standards.
- Description of the containers or cylinders, specifying the capacity, filling pressure of the container or cylinder, type of material used for the container, and the reference code for the manufacturer and suppliers of the containers.
- Capacity and working pressure, valves and valve assembly, including regulators, tubing and the type of valves, their reference codes, the suppliers and the type of valve outlet connection should be stated. The quality of the valve, regulators and tubing may impact the quality of medicinal oxygen being delivered, and therefore regulatory oversight is recommended for these products.
- The principle of the manufacturing method chosen by the manufacturer should be explained and justified.
- Information to support the compatibility and suitability of medicinal oxygen gas with packaging materials and delivery mechanisms. The following considerations should be made:

  – known compatibility of the materials of the cylinders, including information from literature, standards and long-term use of such materials;
  – use of new materials and their gas compatibility (for example, alloys for container shells, plastic materials for valves);
  – choice of new packaging components, especially the valve connections, the most fragile part of the cylinder, especially in terms of airtightness;
  – container filling at a pressure not previously used in the medicinal field for this type of container, with justification required to demonstrate safety, suitability and compliance with regulatory standards;
  – suitability of the container capacity;
  – diversity of cylinders used, if they have quite different capacities and dimensions and are of different materials.

- Information and reports of process validation and evaluation of manufacturing process.
- Control of quality:
  - specifications;
  - details and information on analytical methods or procedures;
  - batch analyses.
- Labels and labelling requirements: the NRA must provide the labelling requirements for a medicinal oxygen on its final use containers, including a warning statement concerning the use of the medicinal oxygen and appropriate directions and warnings concerning storage and handling. The label should include at least the following:
  - product name;
  - dosage form: medicinal oxygen;
  - name of active substances;
  - purity, pressure, temperature (for liquid oxygen);
  - batch or lot number;
  - manufacturing date;
  - shelf-life, where applicable (for example, oxygen filled in cylinders for commercial supply);
  - storage conditions;
  - name and address of manufacturer;
  - safety label: precaution or warning;
  - pack size (unit, volume).
- Transportable pressure equipment (cylinders), including pressure rating, cylinder material, valve type, fill volume and safety markings.
- Fixed pressure equipment (cryogenic tanks), including tank capacity, maximum allowable working pressure, insulation type and operational safety instructions.
- Post-market surveillance: as part of the application for market authorization, the manufacturer of medicinal oxygen should include a comprehensive post-market surveillance plan. This plan should outline the approach for monitoring product performance and identifying any adverse events once the product is on the market. It should also detail mechanisms for incident reporting

and provide clear guidance on managing any issues related to the use of medicinal oxygen, in line with practices for other medicinal products.

- In the case of cylinder bundles, the material and dimensions, for example, internal diameter used for the loop, should be provided. The outlet valves and the suppliers should be provided.
- For therapeutic claims of medicinal oxygen that go beyond conventional oxygen therapy, clinical trial documentation should be submitted, and, where applicable, prior clinical trial documentation should be provided.

6.1.2 Medicinal oxygen to be placed on the market

Medicinal oxygen intended to be placed on the market must undergo an assessment process to ensure compliance with regulatory requirements. This process can only apply to medicinal oxygen manufactured on a large scale[42] for commercial distribution. Liquid oxygen supplied in bulk, as well as cylinders filled commercially from liquid oxygen sources or by operators of medium- and small-sized standalone PSA, VSA or VPSA plants, fall into this category. The NRA should be responsible for granting market authorization after verifying that the manufacturer has met all regulatory requirements. The authorization process should ensure that the product complies with established quality, efficacy and safety standards before it reaches health care facilities or patients.

6.1.3 Medicinal oxygen exempted from market authorization

Medicinal oxygen produced and distributed solely for use within a health care facility and its related satellite facilities, and not intended for external market distribution, may be exempt from the formal market authorization process. However, such oxygen must still meet specific regulatory requirements set by the NRA. These requirements should ensure that oxygen produced within the health care facility maintains consistent quality, efficacy and safety standards, in line with the needs of patients within the facility. The NRA should clearly define these criteria and enforce compliance to safeguard patient health and safety.

6.1.4 Documentation requirements relating to distribution

The delivery of bulk gases in mobile delivery tanks or stationary storage tanks to hospitals should be clearly detailed in the authorization dossier submitted by the manufacturer. Additionally, individually filled oxygen cylinders distributed

[42] Large-scale manufacturing of medicinal oxygen refers to the industrial production of oxygen in high volumes using specialized technologies to supply health care facilities and distributors.

from production sites must include proper documentation and labelling to ensure traceability, safety and compliance. It is important to avail written instructions, procedures and records that document all activities relating to the distribution of medicinal oxygen from the point of production to the point of use. Documentation should be provided to demonstrate that there will be no mixing between the bulk oxygen gas and any gases originating from other manufacturers in the storage tanks intended for the bulk gas.

NRAs should ensure that manufacturers comply with regulatory requirements for transporting, handling and storing bulk medicinal oxygen. This includes validating storage systems, using proper equipment and implementing measures to prevent cross-contamination.

The subsequent distribution or dispensing of medicinal oxygen by third parties is not typically part of the dossier to be assessed and authorized by the NRA, as the dossier focuses primarily on product quality, safety and efficacy at the time of market authorization. However, it is recommended that the NRA maintain oversight of the distribution process through regular inspections, audits and reporting requirements. This ensures that third-party distributors adhere to the necessary quality standards and regulations, safeguarding the integrity of the medicinal oxygen up to the point of use.

7. Considerations for assuring quality of medicinal oxygen

Regardless of its source, whether produced by large-scale manufacturing facilities, packaged in different facilities, delivered in various formats, or produced on-site at health facilities, the quality and purity or concentration of medicinal oxygen must remain consistent and fulfil the regulatory requirements. This consistency is vital to ensure that all oxygen delivered to patients meets the necessary therapeutic specifications, minimizing the risk of adverse events or inefficacy. The NRA should establish regulatory standards to ensure that medicinal oxygen remains free from contaminants and maintains the required purity levels throughout its delivery and use. It is recommended that the NRA ensure that medicinal oxygen complies with the specification for the assay of the gas as well as the limits for impurities, as specified in the monograph of *The International Pharmacopoeia* (*4*) and other recognized pharmacopoeias, or national specifications as specified from time to time where harmonized monographs are unavailable. The NRA should ensure that manufacturers comply with all regulatory requirements, including the implementation of rigorous quality assurance measures.

The NRA should carry out an evaluation of scientific evidence, quality control and quality assurance documentation, and manufacturing practices, storage, inventory management and distribution, to confirm the conformance with requirements of safety and quality of medicinal oxygen.

7.1 Large-scale production of medicinal oxygen

The NRA should oversee production of medicinal oxygen at large-scale manufacturing facilities to ensure compliance with GMP principles throughout production activities, including processing, filling, transfilling, purifying, separating, transferring, packaging and distribution. The specification of medicinal oxygen should adhere to stringent regulatory requirements, with appropriate quality control verifications. The WHO guidelines on good manufacturing practices for medicinal gases serve as a key reference in this context (5).

Facilities intending to manufacture, store, supply or distribute medicinal oxygen must hold a valid manufacturing licence issued by the NRA. Additionally, the product should meet all applicable medicine registration requirements, as prescribed by the NRA.

Continuous production and delivery of oxygen to patient bedsides or points of use, whether from cryogenic or PSA processes, do not require batch-to-batch analysis. This exemption is based on the real-time monitoring of process parameters and online quality controls that ensure the consistency of the delivered product. NRAs should establish requirements for hospitals to ensure compliance with safety and quality standards of oxygen distributed in piping networks. These requirements should include routine testing and monitoring of oxygen at regular intervals based on perceived risk. Documentation of testing results, maintenance logs, flow rates and corrective actions should be maintained.

7.2 Licensing and inspection of establishments for medicinal oxygen

NRAs should establish and enforce requirements for licensing and conducting risk-based inspection of establishments involved in the production, import and wholesale of medicinal oxygen. Such requirements should enhance compliance with best practices in GMP and GSDP as essential components of the medicine regulatory framework.

Manufacturing licence. A manufacturing licence for medicinal oxygen should be mandatory, irrespective of:

- the method of oxygen production (for example, cryogenic distillation or PSA);
- the production scale (for example, small, medium, large);
- the mode of distribution (for example, cylinders);
- the end users (for example, patients).

Inspection and oversight. The NRA should conduct an inspection before issuance of the licence and should continue with regular inspections of establishments to verify their compliance with GMP and GSDP standards.

This includes ensuring that manufacturing processes, storage, and distribution systems meet the required quality and safety standards.

Licensing of importers, wholesalers and transporters. Importers, wholesalers and transporters involved in the distribution of medicinal oxygen should also be licensed by the NRA. Their operations should be in full compliance with GSDP before distribution to end users.

7.3 Post-licensing inspection of medicinal oxygen establishments

The NRA should ensure that licensed establishments undergo regular risk-based inspections to evaluate their continued compliance with QMS, GMP, GSDP and other regulatory requirements. Post-licensing inspections should be periodically conducted by the NRA and when significant changes are proposed to the licensed operations. Significant changes may include alterations to production methods, site changes, new delivery formats or major equipment replacements. These inspections should take place before approval to ensure that any modifications to the production process, storage conditions or distribution practices comply with regulatory standards. The NRA should establish requirements for any production changes and specific notification requirements.

7.4 Storage and distribution of medicinal oxygen

Manufacturing facilities where medicinal oxygen is produced should be equipped with suitable storage areas that meet the necessary security provisions. These facilities must comply with the essential principles of GSDP. The WHO *Good storage and distribution practices for medical products* (*13*) should be referenced to ensure adherence to best practices for the safe storage and distribution of medicinal products.

The NRA should enforce implementation of GSDP guidelines in all licensed establishments through regular inspections and other enforcement activities. The NRA should ensure that personnel responsible for handling, moving or using medicinal oxygen should be adequately trained and deemed competent to perform these tasks under proper supervision. It is essential that written policies and standard operating procedures for the storage and supply of medicinal oxygen are developed and followed, and that their implementation is subject to regulatory oversight.

In cases where medicinal oxygen is supplied into bulk storage tanks, it should be transferred from the tanks to the facility through a medical gas pipeline system (*17*).[43] This pipeline system may include vacuum systems, anaesthetic gas scavenging systems, medical air, and nitrous oxide. The NRA should ensure

[43] When the medical gas pipeline system is defined as a medical device, it should comply with the requirements for its design in accordance with the ISO standard (ISO 7396-1:2016. Medical gas pipeline systems – Part 1: Pipeline systems for compressed medical gases and vacuum) or the equivalent national standard.

that all pipeline systems comply with the required standards and regulations to guarantee the safety and integrity of the oxygen supply. The NRA should ensure that all manual or automatic filling systems, including equipment used to transport and transfill, comply with the required standards and regulations to guarantee the safety of the oxygen supply. These systems must comply with standards for design, installation, validation, verification and ongoing operational management to ensure safe and efficient distribution of medicinal oxygen (*18*). While medicinal oxygen itself is chemically stable and may not degrade over time, the storage vessels (for example, cylinders) are subject to requalification and certification expiry, which must be clearly indicated on the label.

7.4.1 Medicinal oxygen cylinder storage

Regulatory requirements should be established for the handling and storage of medical gas cylinders, whether used as a primary or secondary source of medicinal oxygen, for example when serving as an alternative to a medical gas pipeline system. When not in immediate use, all gas cylinders should be stored under the conditions highlighted in Table A10.3.

Table A10.3
Storage conditions for medicinal oxygen cylinders

Category	Storage conditions
Designated storage area	Medicinal oxygen cylinders should be kept in a designated storage area appropriately marked with status labels, safety signs and warning notices indicating the gas and any potential hazards. There should be segregated areas for full, empty and quarantined cylinders indicating the gas and any potential hazards, in accordance with ISO 7396-1 or equivalent local standards.
Ventilated and safe environment	The storage area should be clean, well ventilated, and away from any sources of ignition, excessive temperatures, flammable materials, or incompatible products (such as food), to minimize safety risks.
Security and access control	The storage location should be secure, with access strictly controlled to prevent unauthorized handling and to ensure the safety of personnel.
Clean and protected storage	Cylinders should be stored under cover, ensuring that the environment is clean, dry and suitable for use in a medical setting, while still maintaining proper ventilation. They should be kept secured with racks or chains and fitted with valve protective devices such as valve caps or valve guards. Cylinders should be inspected regularly for damage, corrosion and valve integrity, and requalified as per the manufacturer's and regulatory requirements.

7.5 Regulatory inspections

Regulatory inspections are essential to verify that medicinal oxygen establishments comply with applicable standards, thereby ensuring that patients receive safe and high-quality medicinal oxygen. The NRA should establish a risk-based inspection system, including development of guidelines and standard operating procedures for internal and external stakeholders. The system should ensure that all entities involved in manufacturing, distribution or importation adhere to regulatory requirements and maintain the integrity of the medicinal oxygen supply chain.

7.5.1 Inspection of medicinal oxygen manufacturers

The NRA should carry out risk-based inspections of medicinal oxygen manufacturers at various stages of the product life cycle to ensure consistency in quality and purity. Additionally, the NRA should require manufacturers to implement strong quality control systems to ensure that medicinal oxygen consistently meets established regulatory requirements. For domestic manufacturers, this encompasses a comprehensive site inspection, including assessment of the QMS and verification of compliance with GMP standards. For overseas manufacturers, the NRA should ensure that a valid GMP compliance certificate is issued and recognized, in accordance with international standards. Where necessary, the NRA should conduct its own inspection of foreign manufacturing sites. Where possible, the NRA may rely on certificates issued by regulatory authorities recognized under mutual recognition or reliance agreements, provided appropriate verification is conducted.

Inspection outcomes should inform decisions related to market authorization, licensing of manufacturing and storage sites, and ongoing regulatory monitoring. The NRA may reference the Pharmaceutical Inspection Co-operation Scheme aide-memoire on inspection of medicinal gases (*19*) and the *WHO good manufacturing practices for medicinal gases* (*5*) for inspections to guide this process.

7.5.2 Inspection of distributors

The NRA should ensure that every importer and wholesaler responsible for placing medicinal oxygen on the market follows GSDP (*13*). This includes ensuring that medicinal oxygen is stored, distributed and handled appropriately under controlled conditions (for example, adequate ventilation, temperature management, and segregation of empty and full cylinders). The NRA should carry out inspections of distributors to verify their adherence to GSDP standards and to ensure that the integrity of the product is maintained throughout the distribution process, and should carry out spot checks for quality of the product.

7.6 Importation of medicinal oxygen

When medicinal oxygen is imported, the manufacturer in the country of origin is responsible for holding the market authorization or may authorize a representative to act on their behalf. The authorized party should also ensure that post-market surveillance obligations are fulfilled in the importing country. The importer is required to apply for market authorization to the NRA, which then evaluates the application to determine whether it complies with national regulations. The NRA should prescribe the designated ports of entry for importation or exportation of medicinal oxygen.

The NRA should follow the standard process applicable to imported medicinal products, which includes the following:

- assess documents for market authorization and, where appropriate, rely on or recognize approvals issued by trusted regulatory authorities or WHO-listed authorities;
- conduct a site inspection to assess and license the premises of the importer, ensuring that the facility meets the required regulatory and safety requirements for handling and distribution of medicinal oxygen;
- verification of importation documents to ensure compliance with applicable regulations and issuance of import authorization;
- Inspection and sampling[44] of medicinal oxygen consignments at the designated ports of entry.

7.7 Quality control testing

The NRA should ensure that manufacturers are conducting batch-specific quality control tests or continuous online analysis for each batch of medicinal oxygen before release for distribution or consumption (including assay and impurity analysis) and follow internationally recognized laboratory control requirements. Any deviations from written specifications, standards, sampling plans, test procedures or other quality control mechanisms should be appropriately recorded and justified. The NRA should ensure that medicinal oxygen, which is derived from ambient air using various oxygen production technologies including PSA and cryogenic processes, is subject to stringent quality control testing to meet national and international standards. These standards should encompass quality, safety, specification, purity, packaging, labelling and identification of the product.

[44] NRAs should develop guidelines and standard operating procedures to guide inspection activities at ports of entry. Sampling and testing may include verification of identity, purity, and compliance with recognized pharmacopoeia standards .

Internationally recognized pharmacopoeias, such as *The International Pharmacopoeia*, the *United States Pharmacopeia* and the *European Pharmacopoeia*, provide specific monographs for medicinal oxygen. The NRA should ensure that manufacturers refer to and comply with any of those standards and perform batch-specific quality control testing before release for medical use, including testing for purity, contaminants, and compliance with monograph standards.

7.7.1 Important factors for quality control testing

To ensure the quality and safety of medicinal oxygen, the NRA should adopt the following roles and responsibilities to guide manufacturers in meeting quality control standards.

1. Medicinal oxygen purity levels

- The NRA should provide clear guidance on the required oxygen specifications based on the production method used – whether cryogenic or PSA systems or other methods. The NRA should ensure that medicinal oxygen purity levels or oxygen concentration (% O_2)[45] meet internationally recognized pharmacopoeia requirements. This purity can vary depending on the production method, using either cryogenic technology or PSA plants. Medicinal oxygen must meet extremely high purity standards, with specified minimum oxygen content and maximum allowable levels of contaminants, such as moisture, oil, carbon dioxide and carbon monoxide.
- For oxygen produced via cryogenic distillation process, the NRA should ensure alignment with the requirement of *The International Pharmacopoeia* of a minimum of 99.5% v/v O_2.
- For PSA-produced oxygen, the NRA should adopt and promote WHO's interim guidance of 93% ± 3%[46] v/v O_2 purity as the minimum acceptable standard (*20*).
- Regulatory oversight should include validation of production methods and regular testing to confirm that oxygen meets the required specifications.

2. Trace contaminants

The NRA should require manufacturers to implement comprehensive quality control systems to detect and limit trace contaminants, such as hydrocarbons,

[45] Oxygen purity or concentration (% O_2) refers to the amount of oxygen present in the final product.

[46] The ± 3% tolerance for PSA applies only under validated production and monitoring conditions.

volatile organic compounds and particulate matter, based on a risk assessment. The quality and safety of medicinal oxygen can be compromised by even trace amounts of impurities. Therefore, robust quality control measures should be implemented to monitor and reduce the levels of potential gas contaminants to acceptable limits. Regulatory guidelines should emphasize routine testing for contaminants associated with PSA, VSA or VPSA and cryogenic oxygen production, including carbon monoxide, carbon dioxide, humidity, sulfur dioxide, nitrogen oxides and oil. The NRA should establish or adopt detailed monographs for PSA-produced oxygen to ensure all relevant contaminants are identified and controlled.

3. Microbial contamination

The NRA should ensure that manufacturers maintain appropriate design and hygiene standards for oxygen supply and storage systems to prevent microbial contamination. The NRA should issue guidance requiring manufacturers to implement practices such as regular cleaning, sanitization and microbiological monitoring.

Regulatory inspections should include assessment of manufacturers' compliance with microbial control procedures.

7.7.2 Quality control processes

NRAs are encouraged to enforce national and internationally recognized quality standards, such as those outlined in *The International Pharmacopoeia*, the *United States Pharmacopeia* and the *European Pharmacopoeia*, or WHO guidelines on medicinal oxygen production. The manufacture of medicinal oxygen should involve procedures to test, monitor and verify quality and consistency of production operations. This may include in-process control by continuous monitoring of critical factors such as oxygen purity, flow rates and pressure levels. The NRA should ensure that in-process control tests are defined and followed and that manufacturers maintain regular calibration in accordance with manufacturer instructions and relevant calibration standards (such as ISO 10012) and maintenance for equipment and instruments. The NRA should also require manufacturers to maintain control charts or logs for continuous monitoring and maintain procedures for handling and documenting deviations.

The NRA should set provisions for final product testing to confirm that the medicinal oxygen meets established purity standards and regulatory requirements. The testing should ensure that all necessary specifications are met before each batch or lot of the product is released for distribution.

The NRA should verify availability of written procedures that clearly outline:

- sampling and testing plans that include the number of units per batch that will be sampled and tested;
- test methods and acceptance criteria for sampling and testing;
- corrective actions to be taken if test results fall outside the specifications.

Note: On-site oxygen generation that continuously feeds a directly connected piping distribution system by nature may not be batch tested. However, it should be continuously monitored to ensure that only medicinal oxygen that meets the required quality and purity standards is administered to patients. The sampling and testing should be carried out by continuous on-line monitoring.

On-site oxygen is produced continuously and supplied via a medical gas pipeline system without prior control or release by a responsible person or releasing officer, which is then administered directly to patients; therefore, continuous monitoring of quality and purity is mandatory. This ensures that any out-of-specification product is detected, and the supply can be switched to a backup source.

Quality control at points of use

The NRA should prescribe the requirements for health care facilities to conduct quality control analyses on the medicinal oxygen they receive and distribute within the health care facility supply system. This also includes prescribing the requirements for distribution devices, such as manifolds, which can affect the oxygen's quality and purity. Extensive regulatory oversight should be considered, since even in pressurized systems, leaks may allow contaminants from the environment to enter the system, potentially affecting the purity of the oxygen, as per Graham's laws of diffusion and effusion (*21*).

The WHO monograph on medicinal oxygen (*4*), pharmacopoeias and other internationally recognized tests provide simple testing methods for detecting contaminants in oxygen, whether it is produced through PSA technology or other means. The NRA should recommend that hospitals perform these tests at the point of use to ensure that the oxygen supplied to patients is free of harmful impurities.

The NRA should emphasize that these tests, as an additional quality assurance measure, are essential to ensure the quality of medicinal oxygen provided to patients. The responsibility for conducting these tests lies with the pharmacist at the health care facility. If no pharmacist is available, this responsibility may be assigned to another qualified professional, such as a biomedical engineer, as specified by the hospital.

7.7.3 Risk analysis and control

The NRA should enforce the implementation of a comprehensive risk management strategy to identify potential risks and vulnerabilities in the production, storage and distribution systems of medicinal oxygen. Manufacturers must ensure that any evaluation of risks to oxygen quality is based on sound scientific principles, with risk assessments being documented. To identify potential failure modes, their causes, and associated risks in the oxygen generation process, the NRA should encourage the use of risk analysis methods such as failure mode and effects analysis. The NRA should follow up and verify adequacy of risk mitigation measures implemented.

7.7.4 Control of the finished product

For finished medicinal oxygen, the product specifications should comply with the requirements set out in the relevant pharmacopoeial monographs. The control of the finished product should consist of the control according to the monograph or using validated methods if shown to be equivalent (identity, assay and impurities). Specifications should at least include tests for identity, assay and impurities according to the monograph in the relevant pharmacopoeia, the appearance of the packages, the labelling, and the content (for cylinders and cylinder bundles). The control of the filling batch can be performed during packaging (for example, weight and volume checks) and the control of leak tightness after filling. Controls should also ensure there is no cross-contamination between different gas types during filling operations, especially in multigas facilities.

7.7.5 Stability testing and expiry considerations

Medicinal oxygen has unique stability characteristics. Unless explicitly specified during market authorization, medicinal oxygen does not require a designated shelf-life, meaning manufacturers may not need to maintain an ongoing stability programme. Thus, bibliographic data may be sufficient to demonstrate stability. However, it is essential that containers and closures are verified to be capable of maintaining the gas at the appropriate pressure before being released for distribution (*22*–*24*). Even in the absence of an expiry date, the condition and safety of storage vessels must be verified periodically, including hydrostatic testing for cylinders and pressure monitoring for bulk tanks.

In cases where a manufacturer assigns an expiration date to medicinal oxygen, this must be supported by stability studies. If an expiration date is provided for a batch, manufacturers should establish, document and implement a stability testing programme. Conversely, if no expiration date is assigned to the medicinal oxygen, supporting stability studies are not required (*10*).

7.7.6 Cylinders and pressure regulators

The cylinder or containers and pressure regulators should comply with the specifications outlined in the existing national or international (ISO) standards concerning equipment intended to contain and deliver gases. The certificates of compliance with the applicable standards should be provided by the manufacturer. The NRA may verify that the cylinders, manifolds and regulators have undergone appropriate testing, inspection and certification prior to use or distribution. NRAs should also require that cylinders be evacuated or purged (*25*) to a specified pressure before refilling to prevent post-process contamination. NRAs should require documentation of periodic inspection, pressure testing and recertification of cylinders in accordance with ISO 9809 (*22*) or applicable local standards.

8. Post-authorization requirements

8.1 Quality monitoring

The NRA should ensure that manufacturers conduct continuous monitoring of registered or listed medicinal oxygen through post-market surveillance activities. Manufacturers, importers and distributors are required to implement a systematic product complaint handling procedure and maintain appropriate records. Any quality or safety concerns related to medicinal oxygen should be promptly reported to the NRA. Furthermore, the NRA should ensure that these stakeholders have a robust procedure in place to address identified issues, including corrective and preventive actions.

8.2 Vigilance system for medicinal oxygen

The NRA shall be responsible for ensuring that an appropriate vigilance system is established to monitor medicinal oxygen, and appropriate action is taken when necessary. The system and procedures in place should be adequate for monitoring, handling, evaluating and providing feedback to reporters or health care providers regarding adverse events. The NRA should ensure that adverse events are reported within the stipulated timelines, and the minimum required information for reporting adverse events includes an identifiable reporter, an identifiable patient, the adverse reaction, and a suspect product (*26*).

Note: Medicinal oxygen-related adverse events are rare but can be serious (examples: contamination, pressure failure, equipment interface issues).

8.3 Post-approval changes or variations

The NRA should be alerted to any changes to the approved medicinal oxygen establishment or product. Such variations may be classified as major, minor or notifications (27), with criteria for each category defined according to national requirements. The NRA must establish clear requirements for handling variations or changes, consistent with provisions outlined in the national medicinal law and relevant regulatory guidelines. Provisions should be put in place to ensure that the approval process for variations does not disrupt the supply chain of medicinal oxygen, particularly in health facilities. For instance, production should be allowed to continue while the NRA reviews minor variations.

8.4 Renewal of licence to manufacture medicinal oxygen

The renewal of registration by the NRA should align with the applicable requirements for the renewal of medicinal products. This process provides an opportunity to assess whether the manufacturer has fulfilled all commitments and to ensure the consistency of the approved product, including verification of all approved variations and ongoing compliance with GMP and QMS requirements.

8.5 Withdrawal or suspension of approval or licence to manufacture medicinal oxygen

In situations where a manufacturer wishes to withdraw an approved medicinal oxygen product, the NRA may deregister or delist the product or suspend its approval or manufacturing licence. This action may also apply to any medicinal oxygen product deemed approved under medicinal law, if deemed necessary by the NRA.

Any withdrawal or suspension for reasons other than commercial considerations should be well documented and appropriately justified, where applicable.

9. On-site production of medicinal oxygen

Oxygen generator plants within health care facilities may produce medicinal oxygen for direct delivery through the hospital's piping system, supported by a backup manifold. These systems should be subject to regulatory oversight to ensure compliance with safety, quality and performance standards. The piped oxygen will reach an area valve service unit and be delivered to the patients' bedside outlets. Oxygen produced from on-site generation plants can also be compressed to fill high-pressure medical gas cylinders. Extreme care should be taken due to risks of fire and contamination.

Production facilities and equipment for on-site oxygen generation should be designed to meet logistical and environmental needs, especially where conventional oxygen supply chains are unreliable. Oxygen-generating plants rely on molecular sieve technology (zeolites) to selectively adsorb and discharge nitrogen, allowing oxygen to concentrate for downstream use. While there are diverse types of oxygen generator plants (for example, PSA, VSA, VPSA) they operate on the principle of pushing pressurized air through a molecular sieve bed for nitrogen removal. NRAs should ensure that health care facilities operating oxygen generator plants possess adequate technical knowledge of their operation and comply with applicable safety standards and regulatory requirements.

The NRA should verify that timely servicing of PSA technology is carried out according to the manufacturer's instructions. Regular maintenance and timely replacement of components, including valves and filters, are crucial for ensuring the optimal performance and safety of the oxygen production and distribution systems.

Additionally, the NRA should enforce requirements for specialized storage and transport systems, such as cylinders and cryogenic tanks for compressed or liquid oxygen, made available for use in health care facilities. The NRA should verify that medicinal oxygen produced by these plants contain no less than 93% ± 3% oxygen derived from ambient air (*28*).

Specific considerations for on-site oxygen production are as follows.

- Production may be exempted from formal market authorization procedures under the medicines law, but still subject to technical authorization and NRA oversight.
- Risk-based QMS and GMP are required, tailored to a different scale compared to industrial manufacturing. The NRA should establish the requirements for a quality management system for on-site production of medicinal oxygen, selecting applicable elements from GMP standards.
- The responsibility for oversight lies with the (hospital) pharmacist.
- Production is not intended for commercial scale or placement on the market.
- Production is demand driven, dependent on the specific need of the health care facility. The health care facility involved should define the quantity of medicinal oxygen to be supplied from on-site production for the facility.
- The NRA should oversee production of medicinal oxygen within health care facilities, including the issuance of manufacturing authorization. On-site production premises should be subject to inspection prior to granting authorization, in accordance with applicable regulatory requirements.
- Where PSA, VSA or VPSA supply sources are used, each plant must be provided with a reliable source of electricity that can maintain

supplies even under single fault conditions. A backup electricity generator may be required to maintain supplies.

- Where supply sources are installed at remote locations, a reliable independent electricity supply source is required for each plant location.
- Inadequate sizing of oxygen generator plants can result in reduced medicinal oxygen purity when the demand exceeds the plant's designed flow rate.
- Continuous analysis: as the medicinal oxygen produced by the oxygen generator plants is not batched and released prior to it being fed to the medicinal gas pipeline system, the product must be continuously analysed for assay and for any contaminant specified by the relevant pharmacopoeia monograph or potential contaminant identified during the initial risk assessment.

9.1 Regulatory framework for the on-site production and oversight of medicinal oxygen: roles, activities and compliance

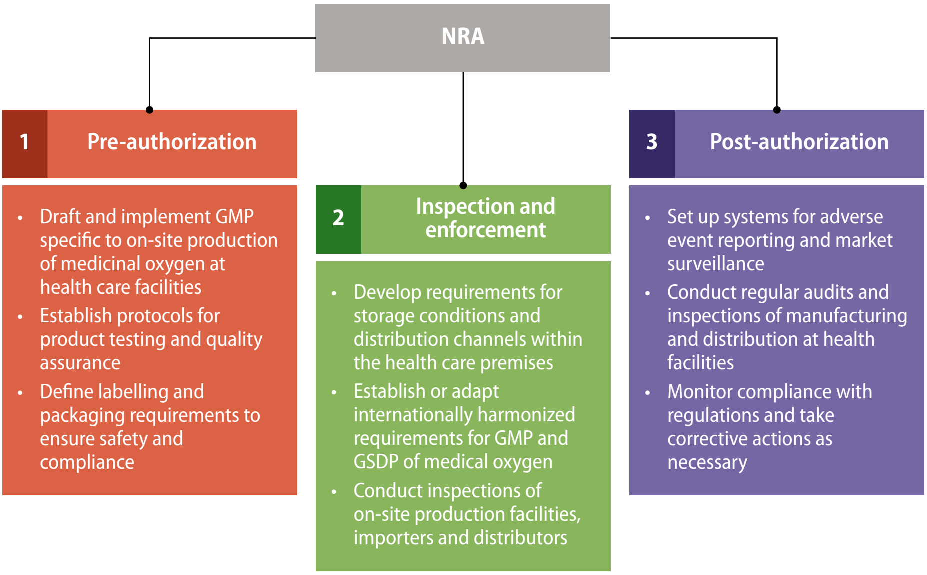
The NRA should establish the framework and activities to be regulated within the supply chain for medicinal oxygen produced on site, outlining the activities that require regulation at each stage (Fig. A10.2):

- pre-authorization activities: manufacturing processes, quality control measures, testing protocols, labelling, packaging and storage;
- inspection and enforcement: GMP and GSDP requirements for on-site production;
- post-authorization activities: monitoring, reporting of adverse events and surveillance.

Furthermore, the NRA should establish compliance teams within the NRA, including regulatory, quality assurance and other technical experts responsible for overseeing these activities. These teams should also coordinate closely with technical teams within hospitals to ensure effective implementation.

Fig. A10.2

Overview of activities in the regulation of medicinal oxygen across pre-, during, and post-authorization stages



Training and awareness

The NRA should organize training sessions for health care and hospital staff involved in the medicinal oxygen life cycle to ensure they are informed about and comply with the regulatory framework. Training should be competency-based and include simulations or mock inspections. Training programmes should be regularly updated to reflect regulatory changes and integrate best practices.

Quality control and testing

While the NRA should retain oversight of quality control practices, the responsibility for establishing and managing testing laboratories should rest with health care facilities. Each health care facility should be required to implement a reliable real-time testing system to verify the purity, identity and impurities of medicinal oxygen.

Documentation and reporting

The NRA should develop comprehensive documentation requirements for all regulated activities related to on-site production. Systems for regular reporting and record keeping should be put in place to ensure transparency and accountability in the production process.

Monitoring and auditing

The NRA should encourage health care facilities to conduct regular internal audits to ensure ongoing compliance with regulatory requirements. Health care facilities should establish procedures for implementing external audits and inspections conducted by the NRA to verify compliance across the supply chain.

Implementation plan

To ensure seamless and efficient regulation of the on-site production of medicinal oxygen, the NRA should take a proactive role in establishing an implementation plan of actions to be undertaken as part of the implementation to ensure the successful oversight and management of this process. As part of the implementation plan, the NRA should:

- identify and compile a list of health care facilities involved in manufacturing of medicinal oxygen;
- establish a transition period to ensure a smooth implementation of regulations;
- develop specific guidance for QMS (based on GMP) for on-site production of medicinal oxygen;
- create requirements and a simplified procedure for assessing the on-site production of medicinal oxygen (see Appendix 2);
- develop an inspection strategy for both manufacturing and distribution sites of medicinal oxygen;
- develop an enforcement strategy.

9.2 Responsibilities of the qualified personnel designated by the health care facility

In health care facilities where medicinal oxygen is produced, the NRA should recommend that the qualified personnel (as determined by the facility's management) are accountable for the quality and safety of the oxygen administered to patients. The NRA should oversee the responsibilities of these qualified personnel. The designated qualified on-site personnel are responsible for:

- maintaining the quality management system and ensuring adherence to GMP for on-site medicinal oxygen production;
- supervising production operations at the health care facility and ensuring compliance with the minimum recommended principles outlined in the *WHO good manufacturing practices for medicinal gases* (5);

- ensuring that the supply of medicinal oxygen from the on-site PSA plant complies with GSDP;
- overseeing quality control requirements to ensure compliance with the relevant pharmacopoeia and national standards;
- establishing testing frequency and quality control procedures to confirm that the oxygen administered to patients meets required quality and purity standards;
- defining appropriate testing frequency to prevent and detect gas contamination;
- ensuring the suitability of oxygen supplied via the medical gas pipeline system by conducting routine inspections of the piping system and performing online testing prior to patient administration;
- maintaining full traceability of oxygen produced on site and administered to patients;
- investigating any reported adverse events related to on-site medicinal oxygen.

References

1. The selection and use of essential medicines 2023: web annex A: World Health Organization Model List of Essential Medicines: 23rd list (2023). Geneva: World Health Organization (https://iris.who.int/handle/10665/371090). Licence: CC BY-NC-SA 3.0 IGO.
2. WHO health topics: oxygen [website]. Geneva: World Health Organization (https://www.who.int/health-topics/oxygen#tab=tab_1, accessed 1 March 2026).
3. Medical gases [website]. British Compressed Gases Association (https://bcga.co.uk/topics/medical-gases/, accessed 1 March 2026).
4. Medicinal oxygen (oxygenium medicinalis). In: The International Pharmacopoeia – 13th edition [online database]. Geneva: World Health Organization; 2025 (https://digicollections.net/phint/202502/index.html#p/home, accessed 1 March 2026).
5. WHO good manufacturing practices for medicinal gases. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fifty-sixth report (Annex 5). Geneva: World Health Organization; 2022 (WHO Technical Report Series, No. 1044; https://iris.who.int/handle/10665/365397). Licence: CC BY-NC-SA 3.0 IGO.
6. WHO good manufacturing practices for pharmaceutical products: main principles. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: forty-eighth report (Annex 2). Geneva: World Health Organization; 2014 (WHO Technical Report Series, No. 986; https://iris.who.int/handle/10665/112733).
7. WHO good distribution practices for pharmaceutical products. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: forty-fourth report (Annex 5). Geneva: World Health Organization; 2010 (WHO Technical Report Series, No. 957; https://iris.who.int/handle/10665/44291).

8. WHO Global Model Regulatory Framework for medical devices including in vitro diagnostic medical devices. In: WHO Expert Committee on Biological Standardization: seventy-sixth report (Annex 3). Geneva: World Health Organization; 2023 (WHO Technical Report Series, No. 1045; https://iris.who.int/items/fa7d2f4e-a535-46dc-bfa1-5db76ff441ea). Licence: CC BY-NC-SA 3.0 IGO.
9. Guidelines on import procedures for pharmaceutical products. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations, thirty-fourth report (Annex 12). Geneva: World Health Organization; 1996 (WHO Technical Report Series, No. 863; https://iris.who.int/handle/10665/41849).
10. Medicinal gases: pharmaceutical documentation (including recommendation on non-clinical safety requirements for well-established medicinal gases): scientific guideline [website]. European Medicines Agency (https://www.ema.europa.eu/en/medicinal-gases-pharmaceutical-documentation-including-recommendation-non-clinical-safety-requirements-well-established-medicinal-gases-scientific-guideline, accessed 1 March 2026).
11. Product safety and market surveillance [website]. European Commission (https://www.consilium.europa.eu/en/policies/product-safety-market-surveillance/, accessed 20 March 2026).
12. WHO post-market surveillance (PMS) for medicines [website]. Geneva: World Health Organization (https://www.who.int/teams/regulation-prequalification/regulation-and-safety/pharmacovigilance, accessed 1 March 2026).
13. Good storage and distribution practices for medical products. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fifty-fourth report (Annex 7). Geneva: World Health Organization; 2020 (WHO Technical Report Series, No. 1025; https://iris.who.int/handle/10665/331814). Licence: CC BY-NC-SA 3.0 IGO.
14. Oxygen sources and distribution for COVID-19 treatment centres: interim guidance, 4 April 2020. Geneva: World Health Organization; 2020 (https://iris.who.int/handle/10665/331746). Licence: CC BY-NC-SA 3.0 IGO.
15. World Health Organization and United Nations Children's Fund. WHO–UNICEF technical specifications and guidance for oxygen therapy devices. Geneva: World Health Organization; 2019 (https://iris.who.int/handle/10665/329874). Licence: CC BY-NC-SA 3.0 IGO.
16. Good regulatory practices in the regulation of medical products. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fifty-fifth report (Annex 11). Geneva: World Health Organization; 2021 (WHO Technical Report Series, No. 1033; https://iris.who.int/handle/10665/340323). Licence: CC BY-NC-SA 3.0 IGO.
17. ISO 7396-1:2016. Medical gas pipeline systems – Part 1: Pipeline systems for compressed medical gases and vacuum. International Organization for Standardization; 2016 (https://www.iso.org/standard/60061.html, accessed 2 March 2026).
18. Committee for Medicinal Products for Human Use. Guideline on medicinal gases: pharmaceutical documentation (including recommendation on non-clinical safety requirements for well established medicinal gases). European Medicines Agency; 2008 (CPMP/QWP/1719/00 Rev 1; https://www.ema.europa.eu/en/documents/scientific-guideline/guideline-medicinal-gases-pharmaceutical-documentation-including-recommendation-non-clinical-safety-requirements-well-established-medicinal-gases-revision-1_en.pdf, accessed 1 March 2026).
19. PIC/S aide-memoire on inspection of medicinal gases. Pharmaceutical Inspection Co operation Scheme; 2007 (https://picscheme.org/docview/3034, accessed 2 March 2026).
20. Technical specifications for pressure swing adsorption (PSA) oxygen plants: interim guidance, 8 June 2020. Geneva: World Health Organization; 2020 (https://iris.who.int/handle/10665/332313). Licence: CC BY-NC-SA 3.0 IGO.

21. Graham's laws of diffusion and effusion. LibreTexts: Chemistry, section 2.9 (https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Map%3A_Physical_Chemistry_for_the_Biosciences_(Chang)/02%3A_Properties_of_Gases/2.09%3A_Graham%27s_Laws_of_Diffusion_and_Effusion, accessed 2 March 2026).
22. ISO 9809-1:2019. Gas cylinders – Design, construction and testing of refillable seamless steel gas cylinders and tubes – Part 1: Quenched and tempered steel cylinders and tubes with tensile strength less than 1 100 MPa. International Organization for Standardization; 2019 (https://www.iso.org/standard/70377.html, accessed 2 March 2026).
23. ISO 7866:2012. Gas cylinders – Refillable seamless aluminium alloy gas cylinders – Design, construction, and testing. International Organization for Standardization; 2012 (https://www.iso.org/standard/37438.html, accessed 2 March 2026).
24. ISO 11119-1:2020. Gas cylinders – Design, construction and testing of refillable composite gas cylinders and tubes – Part 1: Hoop wrapped fibre reinforced composite gas cylinders and tubes up to 450 l. International Organization for Standardization; 2020 (https://www.iso.org/standard/75815.html, accessed 2 March 2026).
25. EudraLex: The Rules Governing Medicinal Products in the European Union. Volume 4: Good manufacturing practice: medicinal products for human and veterinary use. Annex 6: Manufacture of medicinal gases. Brussels: European Commission; 2010 (https://www.gmp-compliance.org/files/guidemgr/2009_07_annex6.pdf, accessed 3 March 2026).
26. Post-approval safety data management: definitions and standards for expedited reporting. ICH Harmonized Tripartite Guideline E2D, current step 4 version, November 2003. Geneva: International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use; 2003 (https://database.ich.org/sites/default/files/E2D_Guideline.pdf, accessed 3 March 2026).
27. WHO general guidance on variations to multisource pharmaceutical products. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations: fiftieth report (Annex 10). Geneva: World Health Organization; 2016 (WHO Technical Report Series No 996; https://iris.who.int/handle/10665/255338). Licence: CC BY-NC-SA 3.0 IGO.
28. WHO technical specifications for health facility based medicinal oxygen systems. Geneva: World Health Organization; 2024 (https://iris.who.int/handle/10665/380072).

Appendix 1. Manufacturing of medicinal oxygen

Large- scale and on-site manufacturing and distribution of medicinal oxygen

Medicinal oxygen, like all medical gases, is a distinct product. It differs from medicinal products in several aspects, including its starting material, physical form, primary packaging, physical properties, dosage form, distribution and method of administration. Managing medicinal oxygen can take place on site in health facilities or off site at manufacturing sites. Although different standards may apply depending on the production method and location, both production methods must adhere to strict regulations for quality assurance and undergo quality control verification, which should be conducted by certified, specialized laboratories.

A detailed diagram of the manufacturing process should be presented, together with the controls carried out at the different stages. The production station should be indicated, and the automated packaging system and specifications of the equipment (for example, pumps and balances) should be described and validated. Validation and revalidation of the cylinder-filling process should be performed. Manufacture of medicinal oxygen involves a series of operations, including the division or distribution and filling into dedicated medicinal packaging. Packaging is often automated and may include prior modification to the physical state of the gas, such as vaporization by heating of liquefied gas or compression.

The concentration of medicinal oxygen is possible using multiple methods as described below, which separate nitrogen and oxygen, and sometimes argon and other rare inert gases, from atmospheric air. The different separation methods to obtain concentrated oxygen are as follows.

1. **Cryogenic distillation of ambient air** using an air separation unit to generate liquid oxygen, which liquefies at –183 °C and contains the purity of not less than 99.5% v/v of O^2.[47] Facilities can be equipped with large bulk liquid oxygen tanks that are refilled periodically by a liquid oxygen tanker from a supplier. The liquid oxygen tank supplies a centrally piped system throughout the health facility by self-vaporization, meaning that a power supply is not required.

   Note: This process is conducted off-site, away from the health facility.

2. **Medical ceramic oxygen generator system** developed by the United States National Aeronautics and Space Administration: oxygen is separated through ionization and electrochemical processes. The output is approximately 99.995% oxygen. It uses

[47] Oxygen 99.5% is produced from ambient air by cryogenic distillation, as defined in *The International Pharmacopoeia*.

ceramic membranes with the unique ability to ionize oxygen atoms from ambient air, transport the ions through the solid membrane wall, and recombine them into oxygen molecules on the other side.

3. **Oxygen generator plants (PSA, VSA, VPSA)**. Oxygen generator plants may be used as an alternative source of supply for the medical gas pipeline system for medicinal oxygen, using a plant that is installed and run on the health care site. Since these plants may be the only medicinal oxygen source of supply available to the hospital, their operations must be strictly controlled due to the fact that the oxygen produced is supplied directly to the medical gas pipeline system without being batched and tested before use. Depending on the design of the plant, it produces the medicinal oxygen to the appropriate grade, using zeolites and molecular sieves in adsorption beds to remove the nitrogen and argon from ambient air, producing oxygen at concentrations varying between 90% and close to 100%.

When produced by means of oxygen generator plants through molecular sieves (that is, no liquification), then the minimum standard for medical grade oxygen should be 93% ± 3% (90–96%) purity v/v from ambient air.

There are a few variations of oxygen generator plants involving different separation technologies to obtain concentrated oxygen, as follows.

– Pressure swing adsorption (PSA). This technology separates gases from a mixture under pressure based on their molecular properties and affinity for an adsorbent such as zeolite. Nitrogen is trapped from the air at high pressure and then released when the pressure inside the generator decreases. The process alternates between high and low pressure to regenerate the adsorbent.
– Vacuum swing adsorption (VSA). Similar to PSA, this method also uses a zeolite sieve to separate oxygen from nitrogen. Instead of an air compressor, a vacuum blower is employed, with fewer adsorption vessels and valves. It leverages the adsorbent's higher selectivity under negative pressure.
– Vacuum pressure swing adsorption (VPSA). This is a mixed technology that combines engineering systems of pressure variations from positive (PSA) to negative (VSA) at different stages of the overall process to concentrate oxygen.
  Oxygen generator plants can be used to generate oxygen in health facilities, which is then piped directly to patient bedsides, or can be further compressed to fill high-pressure gas cylinders.

– Home or bedside oxygen concentrators. These are types of medical devices that use the same technology as PSA plants but on a much smaller scale to meet individual patient needs. The typical oxygen concentration is greater than 82%. The flow rate can be adjusted based on patients' needs. These products are not covered in this document.

Large-scale production of liquid oxygen

Bulk liquid oxygen is typically produced on a large scale at manufacturing facilities. This is the most common method for industrial-scale production of oxygen. The process involves separating atmospheric air – composed of approximately 78% nitrogen, 21% oxygen, and 0.9 % argon, with trace amounts (0.1%) of other gases such as carbon dioxide, moisture, carbon monoxide, and inert gases such as neon, helium, krypton and xenon. The gases are separated into their individual components through cryogenic distillation, whereby air is cooled to a very low temperature until it becomes a liquid. Once in a liquid state, the air is separated into its components by taking advantage of their different boiling points. Since nitrogen has a lower boiling point than oxygen, it vaporizes first, leaving behind high-purity liquid oxygen. Bulk liquid oxygen is then stored and distributed for medical use in health care settings.

Additionally, compressed medicinal oxygen is produced by taking bulk liquid oxygen, vaporizing it under pressure, and then filling high-pressure cylinders. The gas can be filled into cylinders or bundles (manifolded cylinders) or delivered through the medical gas pipeline system.

Storage and distribution of medicinal oxygen

Oxygen can be supplied from the production or distribution site to health facilities, either in liquid or in gas form. Liquid oxygen can be stored at the manufacturing facility in vacuum-insulated pressure vessels. The transportation of liquid oxygen must comply with strict national and international regulations prescribed by the NRA for handling of cryogenic pressure vessels.

Oxygen gas can be compressed and stored in cylinders, which are filled at gas manufacturing plants using methods such as cryogenic distillation, PSA, VSA or VPSA. These cylinders are then transported to health care facilities, where they are connected to manifold systems that distribute the oxygen through medical gas piping to bedside terminal outlets. Alternatively, cylinders can be used directly in patient care areas by connecting with a flow meter to control pressure from the cylinder and measure the flow rate of the gas. The use of cylinders often involves transporting them both within facilities and to and from bulk supply depots for regular refilling, which can present logistical challenges and ongoing cost implications.

Cylinders are durable and refillable containers designed to store compressed gases, such as medicinal oxygen, at high pressures. They can store medicinal oxygen pressurized up to 200 bar (2900 psi) but are more typically used at 150 bar (2175 psi) for standard valve cylinders.

Transportation of packaged medicinal oxygen

The NRA should establish requirements for the transportation of medicinal oxygen to ensure that the finished product is delivered in a manner that provides sufficient protection during transit. Transportation activities should comply with GSDP requirements. If medicinal oxygen cylinders are transported in open vehicles, they must be transported in an upright position and be equipped with appropriate plugs and dust covers to safeguard the valves.

Small-scale PSA oxygen production in health care facilities

NRAs should oversee on-site production systems to ensure compliance with applicable regulations. Continuous monitoring is essential to guarantee that the produced oxygen consistently meets safety and quality standards for patient care. On-site PSA systems typically provide oxygen directly to the hospital pipeline system via manifold systems or fill cylinders for later use. All systems operate by passing pressurized ambient air through molecular sieves (for example, zeolites) to selectively remove nitrogen, resulting in concentrated oxygen for medical use.

Storage and distribution of on-site medicinal oxygen in a health care facility

The NRA should be responsible for oversight to ensure that the oxygen produced on site through PSA, VSA or VPSA plants in health care facilities is stored and distributed safely and efficiently. The concentrated oxygen, in gas form, may be delivered directly to the medical gas pipeline system or stored in high-pressure cylinders filled at designated cylinder filling stations. These high-pressure cylinders serve as a storage method for the oxygen, which is then transported to patient bedside outlets or connected to a distribution manifold for further use. The NRA must oversee and regulate these processes to ensure compliance with safety and quality standards.

Regulatory oversight of categories of medical devices for medicinal oxygen production and delivery

Delivery equipment and medical devices serve as an interface to administer oxygen to the patient. Devices intended to support delivery of medicinal oxygen directly to integrated hospital supply systems through either oxygen concentrators or bedside medical gas outlets of a central piped system should be regulated as medical devices according to the applicable law. Also, some equipment, such as medical gas regulators, demand valves, medical gas flow meters and patient monitoring equipment (such as pulse oximeters), must be appropriately classified and regulated as medical devices. NRAs should oversee all types of devices that are integral to the regulation and conditioning of oxygen gas and patient monitoring, ensuring that they meet the required technical specifications to guarantee their safety and performance. The devices fall into two broad categories: non-invasive, which support the patient without penetrating the airway; and invasive, which require internal access to the airway (for example, endotracheal tubes or tracheostomy). The NRA, or any institution responsible for ensuring the safety and performance of these devices, should ensure that medical devices meet the minimum technical specifications required for their safe use in health care settings.[48]

Medical devices must comply with medical device regulations. A facility that wishes to manufacture medical devices (including medicinal oxygen classified as medical device gases) should be registered with the NRA.

Non-invasive delivery devices

Non-invasive medical devices are mostly intended for single-patient use, connected to patient ventilators, and may be reusable – following reutilization measures and lifespan recommended by the manufacturer. Table 1 lists representative examples of such devices.

Table 1
Examples of non-invasive medicinal oxygen delivery devices

Device	Examples
Oxygen source devices	• Concentrator
Oxygen delivery devices	• High-flow nasal oxygen cannula
	• Face mask
	• Mask with reservoir bag
	• Venturi mask
Conditioning, regulation and testing devices	• Flowmeters – Thorpe tube
	• Valves

[48] WHO technical specifications for health facility based medicinal oxygen systems. Geneva: World Health Organization; 2024 (https://iris.who.int/handle/10665/380072).

Table 1 *continued*

Device	Examples
Conditioning, regulation and testing devices *continued*	• Pressure regulators
	• Heated, humidified high flow
	• Oxygen analysers
	• Flow splitter
	• Non-heated bubble humidifier
	• Tubing (for medical gases)
Ventilator devices	• Ventilators
	• High-flow nasal oxygen
	• Continuous positive airway pressure machine
	• Bilevel positive airway pressure device
Manual ventilator device	• Self-inflating resuscitation bag with mask
	• Heat and moisture exchange filter
	• Colorimetric end-tidal CO_2 detector
Patient monitoring devices	• Pulse oximeter (handheld, tabletop and fingertip)
	• Patient monitoring multiparametric (basic, intermediate and advanced)

Invasive medical devices (invasive ventilators)

Invasive ventilators provide long-term respiratory support in intensive care settings. These devices require trained health care personnel and are capable of delivering up to 99.5% oxygen based on prescribed clinical needs. Proper operation includes setting parameters such as volume, pressure and alarm systems.

Appendix 2. Requirements for medicinal oxygen authorization

Submission components	Medicinal oxygen	
	Manufacturing large scale: off site	Manufacturing: on site
Application form	yes	yes
Quality assurance statement	yes	yes
Label and insert	yes	yes (where applicable)
Premise licence	yes	covered under hospital licence
GMP certificate	yes	adapted GMP requirements
Technical dossier	yes	not applicable
Testing results	yes	yes
Risk management plan	yes	adapted to scope
Location plan	yes	adapted to hospital infrastructure
Stability testing	yes	not applicable
Credentials of the pharmacist in charge	yes	yes
Credentials of other qualified personnel	yes	adapted to country's requirements
Other procedures, protocols, records, and reports required by WHO or Pharmaceutical Inspection Co-operation Scheme GMP	yes	as per prescribed GMP requirements

Source: Adapted from: Regulations for registration of medicinal products In: Laws and regulations database of the Republic of China (Taiwan) (https://law.moj.gov.tw/ENG/LawClass/LawParaDeatil.aspx?pcode=L0030057&bp=4).